

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE601 - POWER SYSTEMS ANALYSIS - I

20EE601 / 17EE601 - POWER SYSTEM ANALYSIS AND STABILITY

Duration: 3 Hours

Max. Marks: 100

*Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

2) Assume missing data suitably.

Unit – I

1. a) With neat current oscillogram, write a note on short circuit of a synchronous generator on no-load and constant excitation.
- b) Draw the reactance diagram for one-line diagram of Figure 1b.

G1: 15 MVA, 6.6kV, $x''=12\%$
 T1: 20MVA, 6.6/66kV, $x=8\%$,
 T2: 20MVA, 66/6.6kV, $x=8\%$,
 M1 & M2 : 5MVA, 6.6Kv, $x''=20\%$.

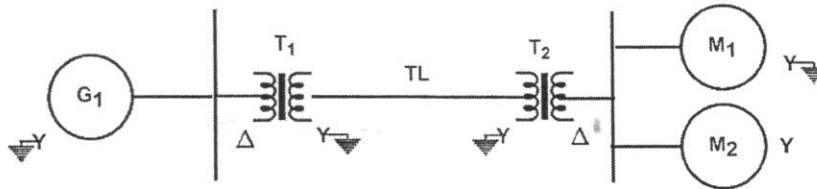


Figure 1b.

2. a) Show that the p.u. impedance of a transformer is same irrespective of the side on which it is calculated.
- b) For the system shown in Figure 2b, the ratings of the various components are:

Generator: 25 MVA, 12.4 kV, 10% subtransient reactance.
 Motor: 20 MVA, 3.8 kV, 15 % subtransient reactance.
 Transformer T1: 25 MVA, 11/33 kV, 8% reactance.
 Transformer T2: 20 MVA, 33/3.3 kV, 10% reactance.
 Line: 20 ohms reactance.

The system is loaded so that the motor is drawing 15 MW at 0.9 leading power factor, the motor terminal voltage being 3.1 kV. Find the sub transient current in generator and motor for a symmetrical fault at a motor bus using Thevenin's method. Choose a base of 25 MVA, 11 kV on generator side.

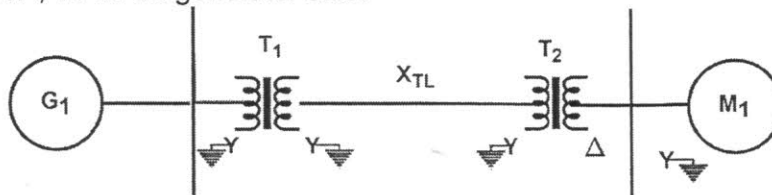


Figure 2b

3. a) Explain doubling effect in a short transmission line by representing it as a series R-L circuit.
- b) Define per unit quantity. Mention any three advantages of per unit system.
- c) Write a note on short-circuit MVA.

P.T.O.

Marks	BT*	CO*	PO*
10	L*2	2	1,2

10	L2	1	1,2
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8	L2	1	1,2
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12	L3	2	1,2
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10	L1	2	1,2
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6	L2	1	1,2
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4	L1	2	1,2
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Unit – II

4. a) Derive an expression for fault current when a double-line to ground fault occurs at terminals of an unloaded generator without fault impedance. Draw the sequence networks to represent the fault.
 b) Draw the positive, negative and zero sequence networks for the power system shown in Figure 4b.

12 L2 4 1

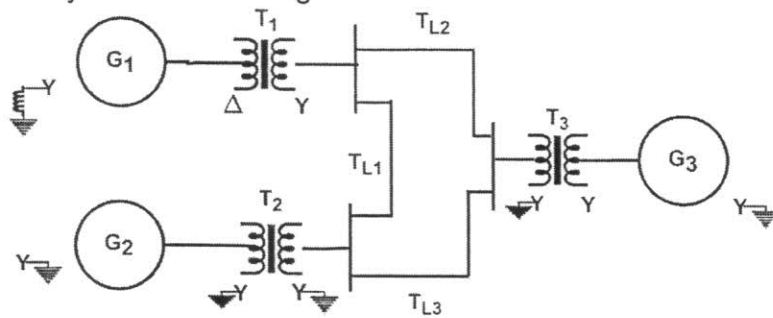


Figure 4b.

Power System Component	Impedance		
	+ve sequence	-ve sequence	Zero sequence
G1, G2, G3	j0.12	j0.08	j0.03
T1, T2, T3	j0.1	j0.1	j0.1
TL1, TL2, TL3	j0.08	j0.08	j0.12

Per unit impedance Z_n in the neutral of $G1=j0.02$ p.u.

8 L2 3 1,2

5. a) A line-to-ground fault occurs at the terminals of an unloaded 3-phase generator. Derive an expression for the fault current.
 b) With the help of the relevant phasor diagrams of voltages, show that there exists a phase shift of positive and negative sequence components in the 3-phase star- delta transformer bank. Assume the HT side to be star connected and LT side to be Delta connected.
6. a) Derive an expression for fault current when a double-line fault occurs at terminals of generator. Draw the sequence networks to represent the fault.
 b) Draw the zero sequence network for transformer with connection i) Y-Y isolated neutrals ii) Y-Y both neutral grounded iii) Y- Δ with neutral Y grounded iv) Δ - Δ .

10 L2 4 1,2

10 L3 3 2

12 L2 4 1

8 L3 3 2

Unit – III

7. a) Explain the Runge-Kutta 4th order Method for the solution of swing equation.
 b) Explain the equal area criterion when a power system is subjected to sudden change in mechanical input.
8. a) Explain the point-by-point method Method for the solution of swing equation.
 b) A 50 Hz, 4 pole turbogenerator rated 350 MVA, 11 kV has an inertia constant $H=7$ MJ/MVA. Find
 (i) K.E. stored in the rotor at synchronous speed.
 (ii) The angular acceleration if the electrical power developed is 200 MW when the input minus rotational losses is 600 kHP.

10 L1 5 1,2

10 L1 5 1,2

12 L1 5 1,2

8 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
